Rebalancing the budget does not change the ranking. When πV instead rebalances the budget at Housing First's drug price (third column), the screen's savings buy more vouchers. πV therefore rises from 6.75 to 6.81 percent, and 58 percent of households gain, but only by 0.05 percent of consumption on average. The gain is small and spread thin, while the loss is concentrated on the users the screen takes out: 96 percent of households with a SUD are worse off, and newborns still lose 1.09 percent of lifetime consumption as adults (since childhood is not counted here). Correcting moral hazard's cost to the budget is still not enough to outweigh the loss of that option to households that choose optimally.

When the drug price is also allowed to clear the drug market (fifth column), newborns still lose 0.74 percent as adults. The change from the third column is the general-equilibrium effect of the drug price and not a correction of moral hazard: the screen lowers demand, the price falls by 4 percent, the cheaper drug draws users back, and more of them are caught and lose their vouchers, which the budget then gives to others.

Treatment First with externalities. Having ruled out moral hazard as a reason for sobriety conditions, I now test, in the three steps set out at the start of Section 6, whether the externalities are a reason to impose them. Each step is a column of Table 5. Here I follow what each step does to drug use, to homelessness and to the children, and give its welfare gain; Section 6.3 then takes each gain apart into the externality and the rest.

With the drug price and πV both held at their Housing First levels (seventh column of Table 5), only the screen acts, as in the second column: drug use falls by 43 percent, homelessness falls by 8 percent, and the share of children in a shelter falls from 1.03 to 0.95 percent. In this partial equilibrium, the parents themselves lose, as they do without the externalities: the screen takes an option from them and gives nothing back, and they lose 1.19 percent of their own lifetime consumption, close to the 1.17 percent of the second column. But in contrast to the model without the externalities, a newborn gains 2.82 percent, because the child gains more than its parent loses. The children of the users the screen deters get more of their parents' consumption and housing, and more of them start adult life without the earnings penalty, while the parent

counts only the share ωc of either. In the aggregate, these budget and earnings externalities, which Housing First leaves in place, come with no trade-off against the childhood homelessness externality: by cutting use, the screen keeps more children out of a shelter than it sends to one by putting their parents out.

Letting πV clear the budget at Housing First's drug price (eighth column) spends the revenue the screen frees on vouchers for households that do not use, or use but are not caught: πV rises to 9.24 percent, drug use falls by 42 percent, homelessness falls by 20 percent, and the share of children in a shelter falls by 19 percent. Now the higher πV makes the parents gain as well, by 2.42 percent of their own lifetime consumption, and a newborn gains 6.45 percent, because the higher πV houses more children. Drug use falls by about as much as with both held. Childhood homelessness falls because the additional spending on vouchers lifts other low-income households of child-rearing age that do not use drugs out of homelessness.

Letting the drug price clear instead, with πV held at its Housing First level (ninth column), lowers the drug price by 6 percent. The revenue the screen frees then goes unspent: drug

use falls by 20 percent, homelessness falls by 4 percent, and a newborn gains 0.98 percent.

Finally, letting both the drug price clear the drug market and πV clear the budget gives the main experiment (last column). Under this configuration, Treatment First cuts drug use by 21 percent, from 8.92 to 7.07 percent of households, and the share of households with a SUD by 19 percent, from 3.83 to 3.08 percent of households. It also raises the share of households with a SUD who are in treatment, from 24.6 to 29.0 percent, because a household with a SUD that is in treatment and does not use passes the screen.39

As users who lose their vouchers leave the programme, it costs less at a given πV . Tax revenue also rises, mainly because more children start adult life with higher productivity, and to a lesser extent because fewer adults are unemployed owing to their SUD. Thus, πV rises from 6.75 to 7.80 percent, by 16 percent, making staying clean of drugs even more desirable. At the same time, taking users out of the drug market lowers the drug price by 5 percent, which draws some households back.

In full general equilibrium, Treatment First also lowers homelessness by 11 percent, from 0.62 to 0.55 percent of households, for three reasons. First, households that stop using spend less on drugs, and thus more of them can house themselves. Second, less SUD means less unemployment and so less homelessness. Third, the vouchers Treatment First revokes from households with a SUD go, through a higher πV , to other low-income households. For households aged 26 to 50, the years a child lives with them, homelessness also falls by 11 percent, with the share of children in a shelter falling from 1.03 to 0.92 percent.

These are long-run outcomes, but the transition does not qualitatively reverse them. One concern is that homelessness and childhood homelessness could rise in the first years after the reform, because the households that already have a SUD face losing their vouchers unless they enter treatment and stay sober, while the number with a SUD falls only as fast as recovery allows. As I show, this concern does not materialise: few households lose their vouchers, and those that stop using spend less on drugs and can afford to house themselves, so neither rises even in the first year (Section 6.4).

A newborn is better off under Treatment First by 2.57 percent of lifetime consumption, and its parent by 0.69 percent of its own. The gain of 2.57 percent is below the 6.45 percent of the second step, where the drug price is held, because the cheaper drug draws users back, which lowers πV , and their use harms their children. The share of children living with a parent who uses drugs falls from 16.35 to 13.24 percent, and the share living with one who has a SUD falls from 5.82 to 4.75 percent.

Of the households that enter adult life at 26, 99.9 percent are better off under Treatment First than they would be in the same state under Housing First (Table 5). This is because an entrant has never had the disorder and holds no voucher, so the screen costs it nothing unless it chooses to use, while a voucher is easier to draw and the drug is cheaper. Across all households, 74 percent are better off and 3 percent worse off, and the rest are essentially unaffected. Finally, even 24 percent of those that have a SUD are better off.

39Under Treatment First a household with a SUD keeps its voucher by entering treatment and stopping. I also ran the experiment in which households with a SUD cannot respond at all: any use and the disorder itself fail the screen, in treatment or not, so treatment cannot help a household keep its voucher, although the government still pays for it. The results

continue to hold: Treatment First then cuts drug use by 29 percent and the share with a SUD by 29 percent, homelessness falls by about as much as in the main experiment, by 11 percent, and a newborn gains at least as much. The results therefore do not depend on households with a SUD entering treatment and stopping.

Table 5: The long run under each policy

No externalities With externalities

Treatment First Treatment First

Housing Both πV Price Both Housing Both πV Price Both First held clears clears clear First held clears clears clear

Adults Drug price, benchmark = 1 1.03 1.03 1.03 0.99 0.99 1.00 1.00 1.00 0.94 0.95 Voucher probability πV 6.75 6.75 6.81 6.75 6.85 6.75 6.75 9.24 6.75 7.80 Drug use 8.97 6.48 6.48 7.47 7.46 8.92 5.08 5.19 7.10 7.07 Carrying a SUD 3.85 2.82 2.82 3.25 3.25 3.83 2.22 2.28 3.09 3.08 Homelessness 0.60 0.57 0.57 0.59 0.59 0.62 0.57 0.49 0.59 0.55 SUD share of the homeless 20.6 15.1 15.1 16.8 16.8 20.2 12.6 11.9 15.9 15.6 In treatment, of those with a SUD 24.7 29.0 29.0 29.0 29.0 24.6 29.3 28.7 29.0 29.0

Children Living with a parent who uses drugs 16.44 12.15 12.14 13.97 13.96 16.35 9.53 9.76 13.29 13.24 Living with a parent who has a SUD 5.85 4.37 4.37 5.01 5.00 5.82 3.43 3.56 4.76 4.75 In a shelter 1.01 0.96 0.95 0.98 0.98 1.03 0.95 0.83 0.98 0.92

Welfare CEV of a newborn, percent of lifetime consumption – −1.17 −1.09 −0.89 −0.74 – 2.82 6.45 0.98 2.57 Share of the newborn's gain that comes in childhood – – – – – – 86 59 102 64 Households entering adult life at 26 better off – 0 1 0 18 – 0 100 5 100 All households better off – 0 58 1 63 – 0 74 1 74 Households with a SUD better off – 0 0 13 13 – 0 17 14 24

Notes: percent unless the row says otherwise; each column is a steady state. The first five columns are Housing First and Treatment First without the externalities, with the child appearing only as an adult. Under Housing First the drug price clears and πV is the benchmark's. Under Treatment First, the drug price and πV are both held at the first column's levels in the second column; the drug price is held and πV clears the budget against the first column's net revenue in the third; the drug price clears with πV held in the fourth; and both clear in the fifth. The CEV in the second to fifth columns is the newborn's value at 26, against the first column. Housing First, the sixth column, is the benchmark with the externalities. Treatment First screens with a detection rate of 20 percent. In the seventh column the drug price and πV are both held at their Housing First levels, in the eighth the drug price is held and πV clears, in the ninth πV is held and the drug price clears, and in the last both clear; when πV is held, the revenue the screen frees goes unspent, and when it clears, πV holds net government revenue at its benchmark level. Children are those living with households aged 26 to 50, the years a child lives at home. The CEV is against the benchmark. A household is better off when its value under Treatment First exceeds its value in the same state under Housing First, with households weighted by the Housing First distribution.

6.3 Where the gain comes from

A newborn's gain under Treatment First is the change in (31), and it has four parts, two of which are externalities. Write ΔC for the change in the child's welfare in childhood, the first term of (31), and ΔA for the change in its welfare in adult life, the second.

Childhood its parent does not count. The parent values the child's childhood at the share ωc; the share $1 - \omega c$ of ΔC is the childhood externality, the budget and childhood homelessness externalities together: the consumption and housing a child gets when its parent spends less on the drug and is less often unemployed.